ENKU R0.1 ELECTRICAL ARCHITECTURE

Base R0.1 production prototype. Pro/frontlight/wireless are deferred to a later hardware revision.

Power

File: power.kicad_sch

MCU + IO

File: mcu_io.kicad_sch

EPD + HV

File: epd_hv.kicad_sch

Connectors

File: connectors.kicad_sch

NOT FABRICATION READY until ERC + exact footprint review

Hierarchical schematic capture baseline

ENKU open-source e-reader

Sheet: /

File: enku-mainboard-r0.1.kicad_sch

Title: ENKU Mainboard R0.1

Size: A4	Date: 2026-10-03	**Rev: R0.1**
KiCad E.D.A. 10.0.7		Id: 1/5

R132: USB-only BQ24072T power-path; AO3401A hard reader switch; TPS63802 3V3. No Dock mux in this Base PCB.
BQ24072T: 4.20V; ISET 2k 1%: 445mA nominal.
EN2/EN1: reset/OFF (high-Z) 00 USB100; 01 USB500; 11 suspend.
GPIO16 EN1; GPIO2 EN2; GPIO38 CE. Boot/USB/thermal/NTC qualification HOLD.

TPS63802 replaces TPS63031 for first-spin headroom: 2A @ 3.3V for VIN>=2.3V, MODE low for power-save, 0.47uH DFE201612E-R47M=P2, 511k/91k feedback, 10uF input and 2x22uF output.

J1 PH2: two-wire PCM pack. JB: separate real pouch NTC103JT-050: 40.2k/33.2k bias, 100k series; nominal4.90-35.01C; R81-open hot model inhibits; bench HOLD.

U3 BQ24072TRGTR

U4 TPS63802DLAR

C2 1uF 10V

C4 1uF 25V

C9 22uF 10V

C10 100nF

C11 22uF 10V

C12 22uF 10V

R14 511k 1%

0.47uH DFE201612E-R47M=P2

R15 91k 1%

R16 R13 100k

100k

SW1 MK-12C01-G015

J1 1S LiPo PCM / PH-2

C5 4.7uF 25V

C6 100nF 25V

C7 22uF 10V

C8 22uF 10V

Q2 AO3401A

R40 100k 1%

JB

External cell NTC / hand solder

R135: real retained cell NTC. Conservative nominal4.90-35.01C window,445mA/8.725h; full-cycle and thermal hardware qualification HOLD.

R8 3.48k 1%

R9 2k 1%

R10 100k

R11 10k

R12 10k

BQ24072T PH2 battery + separate cell NTC + switched 3V3
ENKU
Sheet: /Power/
File: power.kicad_sch

Title: ENKU Base — Power

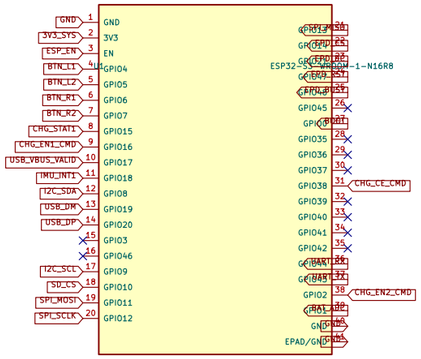
Size: A3 | Date: 2026-10-11 | **Rev: R135 NO FAB**
KiCad E.D.A. 10.0.7 | Id: 2/5

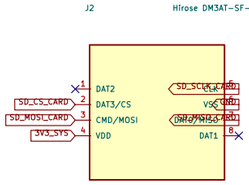
BMI270 is wired in I2C mode: SDO=GND (default address), CSB=3V3, VDD/VDDIO=3V3, INT1 only; unused secondary/OIS pins are NC.

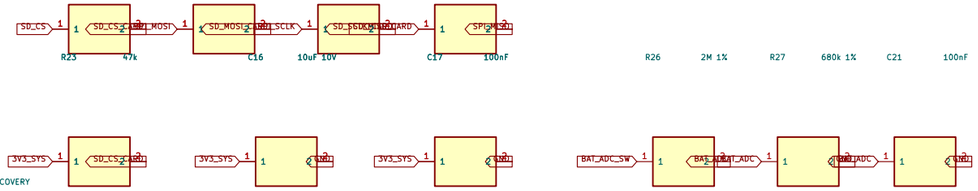
BASE R28: Dedicated PCB — Hall sensor and cover circuitry removed (not DNP): no wireless charging, no frontlight. Separate Pro PCB.

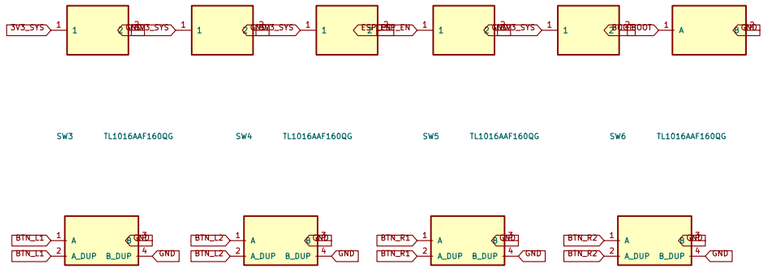
microSD stays on shared SPI with independent CS and OR/22R tuning links. DAT1/DAT2 unused in SPI mode.

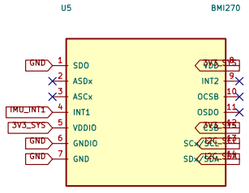
Battery ADC divider 2M/680k draws ~1.6uA at 4.2V; 100nF at ADC node provides acquisition charge reservoir. Validate ADC settling/calibration in firmware.

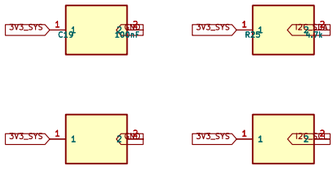


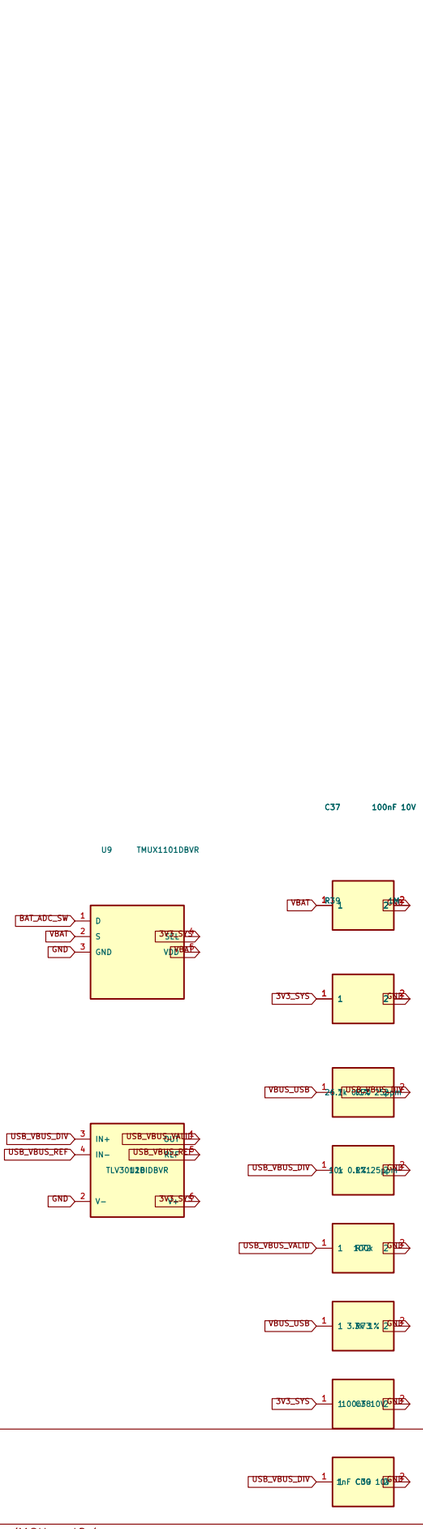












ENKU

Sheet: /MCU + IO/

File: mcu_io.kicad_sch

Title: ENKU R0.1 — MCU + IO

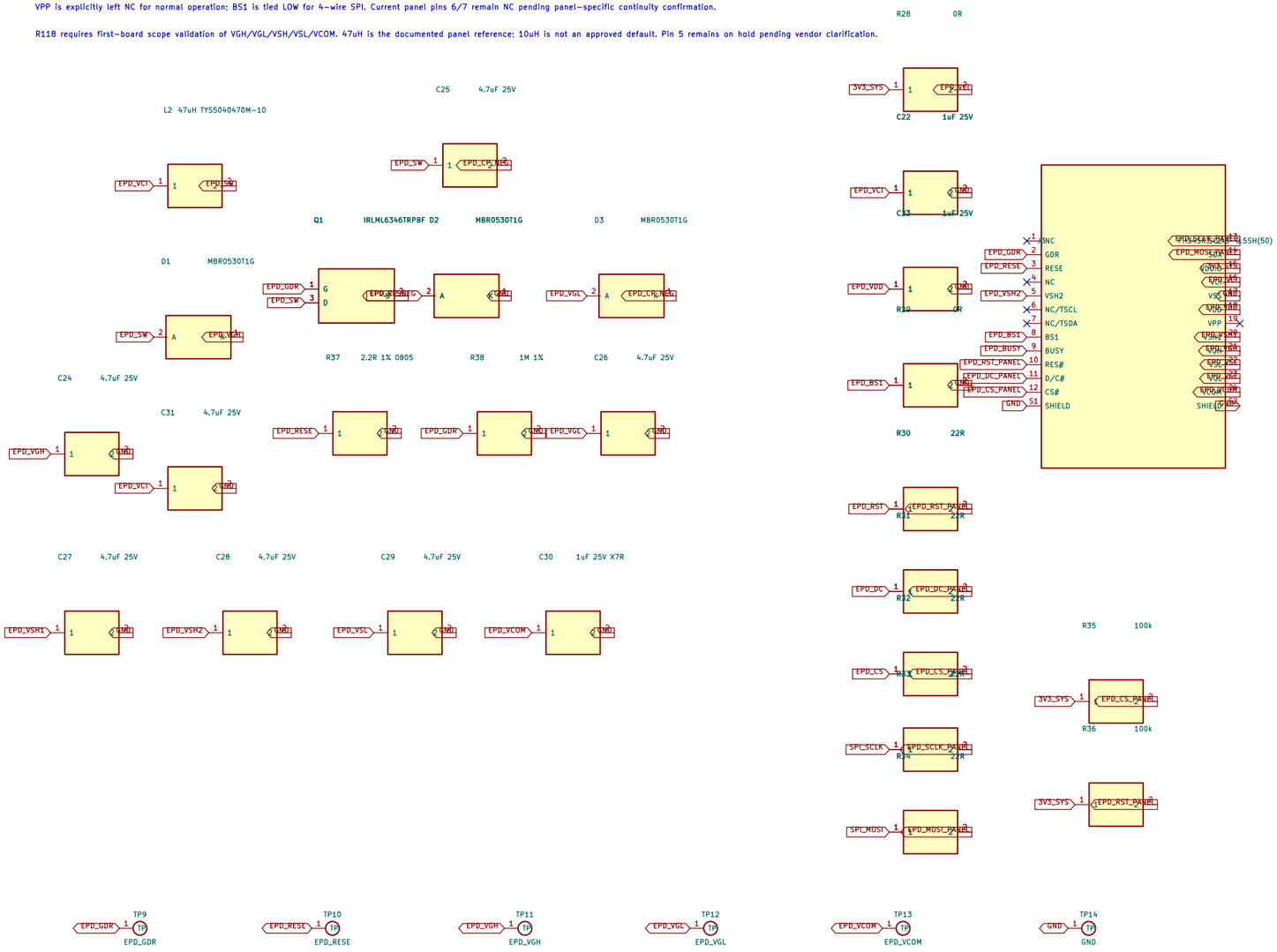
Size: A3	Date: 2026-10-03	Rev: R0.1 discrete capture A
KiCad E.D.A. 10.0.7		Id: 3/5

SSD1677 application circuit captured as discretes. R118 follows GDEY0397T81P supplied datasheet p19: 47uH + NMOS + 3x MBR0530 + 2.2R RESE + 1M GDR pulldown + local 4.7uF input capacitor.

D1: SW -> VGH. C3 flying capacitor + D2/D3 form the negative VGL pump. Q_EPD source is sensed at RESE through 2.2R to GND.

VPP is explicitly left NC for normal operation; BS1 is tied LOW for 4-wire SPI. Current panel pins 6/7 remain NC pending panel-specific continuity confirmation.

R118 requires first-board scope validation of VGH/VGL/VSH/VSL/VCOM. 47uH is the documented panel reference; 10uH is not an approved default. Pin 5 remains on hold pending vendor clarification.



ENKU

Sheet: /EPD + HV/
File: epd_hv.kicad_sch

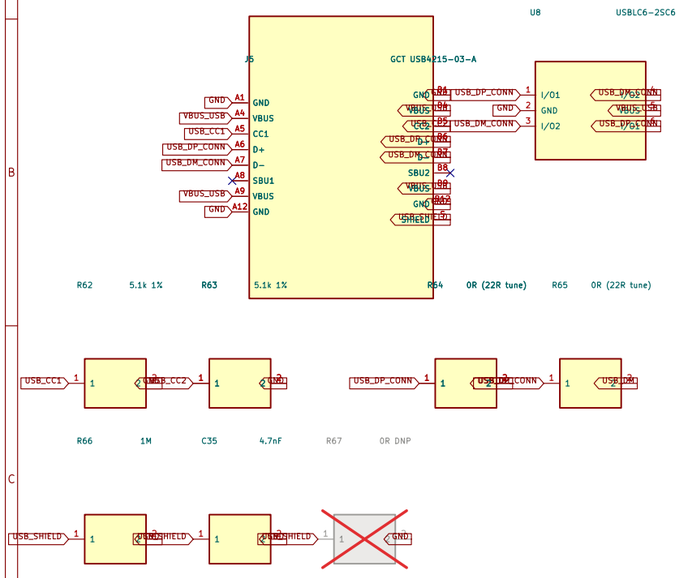
Title: ENKU R0.1 — SSD1677 EPD + HV

Size: A3	Date: 2026-10-03	**Rev: R0.1 discrete capture A**
KiCad E.D.A. 10.0.7		Id: 4/5

USB-C is now captured as the full 16-pin USB2 receptacle: all VBUS/GND pins joined, 5.1k Rd on CC1/CC2, SBU left NC.

D+/D- get an ESD footprint and populated 0R tuning links; the links can be changed to 22R after SI/EMI validation without a respin.

Dock detect retains its 10k series / 100k pulldown / 10nF filter baseline. QI is removed from Base R0.1 and deferred to a later Pro/Wireless revision.





ENKU

Sheet: /Connectors/
File: connectors.kicad_sch

Title: ENKU R0.1 — External connectors

Size: A3	Date: 2026-10-03	**Rev: R0.1 discrete capture A**
KiCad E.D.A. 10.0.7		Id: 6/5